No positive second free Stein discrepancy value

Problem 00000007733: the fixed semicircle target

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The precise printed clause. The source ranges over all atomless distributions supported on $[-2,2]$, asserts minimum 0 at the semicircle law, and then asserts a "next-smallest value." We use its usual meaning: a next *distinct*, strictly positive value. The target is fixed throughout at the unit semicircle, with potential $V_1(x) = x^2/2$. Support means containment in $[-2,2]$; the source imposes no variance-one constraint.

We prove, from actual kernels and their full L^2 infimum, that for every $c > 0$ there is a genuine atomless operator law μ_s with

$$\operatorname{supp}(\mu_s) \subseteq [-2,2], \qquad 0 < \Sigma^*(\mu_s \mid V_1) = 1 - s^2 < c.$$

Thus no positive second distinct value can exist in the printed class. This addresses neither the atomic three-point assertion nor a Jacobi-family classification. Exact-full-support, variance-normalized or stationary-value repairs are outside this claim.

The actual measures and bounded operators. Define the probability measure

$$d\nu(x) = \frac{2}{\pi}\sqrt{1-x^2}\,1_{[-1,1]}(x)\,dx, \qquad \mu_s = (x \mapsto 2sx)_*\nu, \qquad 0 < s \le 1.$$

Lean proves the density integral is 1, atomlessness from absolute continuity, and atomlessness of the pushforward from injectivity. The actual almost-everywhere support bounds give the interval inclusion. The mean is 0 and the second moment is s^2, both computed as native measure integrals. At $s = 1$, change of variables identifies every continuous test integral with the unit Wigner density

$$\int f\,d\mu_1 = \int_{-2}^{2} f(x)\frac{\sqrt{4-x^2}}{2\pi}\,dx.$$

On the native complex Hilbert space $H_s = L^2(\mu_s)$, let $T_s f(x) = xf(x)$. It is constructed as a continuous linear operator using the genuine $L^\infty \times L^2 \to L^2$ multiplication map. We prove

$$T_s = T_s^*, \quad \|T_s\| \le 2, \quad \phi_s(B) = \langle 1, B1\rangle, \quad \phi_s(1) = 1, \quad \phi_s(B^*B) \ge 0.$$

For every $P \in \mathbb{C}[X]$, its native algebra evaluation satisfies $\phi_s(P(T_s)) = \int P(x)\,d\mu_s(x)$. The state is also proved tracial on the actual commutative polynomial operator algebra; no trace assertion on all bounded operators is used.

All polynomial tests, derived from the density. For a complex polynomial P, use the genuine divided difference

$$(\widetilde{\partial}P)(x,y) = \begin{cases} P'(x), & x = y, \\ (P(x) - P(y))/(x-y), & x \neq y. \end{cases}$$

We prove its diagonal derivative convention and measurability. Atomlessness also gives that the diagonal has zero product measure.

Let U_n and T_n be the native Chebyshev polynomials. Actual weighted integrals, using native T_n orthogonality, give

$$\int U_n(x)\, d\nu(x) = \begin{cases} 1, & n = 0, \\ 0, & n > 0. \end{cases}$$

The algebraic convolution identity is proved in every degree:

$$2(x-y)\sum_{i=0}^{n} U_i(x)U_{n-i}(y) = U_{n+1}(x) - U_{n+1}(y).$$

Its actual product integral and the coordinate–U_n integral establish the free integration identity on each U_n. The full complex polynomial sequence is a native basis, so a proved submodule closure extends the result to *every* complex polynomial, not a finite collection of moments. Under the actual dilation pushforward this becomes

$$\int xP(x)\, d\mu_s(x) = s^2 \int (\widetilde{\partial}P)(x,y)\, d(\mu_s \otimes \mu_s)(x,y). \tag{1}$$

All tests are proved to belong to every relevant native L^p space. The all-degree moment recursion is also explicitly derived by Fubini:

$$m_0 = 1, \quad m_1 = 0, \quad m_2 = s^2, \qquad m_{n+2} = s^2 \sum_{i=0}^{n} m_i m_{n-i}, \quad m_n = \int x^n\, d\mu_s(x).$$

The full kernel domain and genuine infimum. We formalize Definition 1.8 of Fathi–Nelson in its one-variable measure form from Section 4. Kernels range over the *entire* native complex Hilbert space $L^2(\mu_s \otimes \mu_s)$. Every $P \in \mathbb{C}[X]$ must satisfy the Stein inner-product identity. Lean's conjugate-linear-first convention is handled by proved integral/conjugation identities. No restriction to constant, symmetric, polynomial, real or finite-dimensional kernels occurs.

Equation (1) proves that the actual constant function $A_s = s^2$ is an admissible kernel. For any admissible kernel A, testing $P = X$ gives $\langle 1, A\rangle = s^2$. Since the constant-one vector has norm 1, Cauchy–Schwarz yields

$$1 - s^2 = |\langle 1, A - 1\rangle| \leq \|A - 1\|_2.$$

The constant A_s attains equality. The set of distances is explicitly proved nonempty and bounded below before native `sInf` is used. Consequently its actual full-domain infimum is

$$\boxed{\Sigma^*(\mu_s \mid V_1) = \inf_{A \text{ Stein kernel}} \|A - 1\|_2 = 1 - s^2.}$$

There is no next distinct value. At $s = 1$ the same actual infimum is 0. Given any $c > 0$, put

$$\varepsilon = \min(c/4, 1/4), \qquad s = 1 - \varepsilon.$$

Then $0 < s < 1$ and

$$0 < 1 - s^2 = 2\varepsilon - \varepsilon^2 < c.$$

The theorem `operator_laws_have_no_positive_next_value` includes the real measure, its atomlessness/support, the actual self-adjoint bounded operator, the normalized positive state, every complex polynomial law identity and the strict discrepancy inequalities. The separate theorem `no_positive_gap_on_admissible_family` gives the logical negation of a positive gap. Since this family lies inside the source's full class, any proposed positive second distinct value for the full class has a smaller positive value inside this family.

Formalization boundary. The one-variable measure/product-measure formulation is the published definition in Fathi–Nelson, Section 4, read together with the all-kernel infimum of Definition 1.8. The proof constructs those actual measures and their native L^2 spaces directly. It also constructs their coordinate operators and proves their complete polynomial distribution identities. A general von Neumann algebra/GNS/tensor-product reconstruction and a general spectral functional-calculus reconstruction are not repeated in Lean. No added axiom substitutes for either reconstruction. No identity between Stein discrepancy and free Fisher information is claimed.

Proof map.

Actual construction	Main declarations
Density, probability and all basis tests	`baseLaw`, `base_U_integral`, `all_base_tests`
Pushforward and all moments	`law`, `law_free_integration_by_parts`, `all_moments_recurrence`
All-kernel infimum and attainment	`IsSteinKernel`, `kernel_norm_lower`, `discrepancy_eq`
Bounded self-adjoint operator law	`coordinateOperator` `coordinateOperator_distribution`

Reproduction and coverage. The complete project pins Lean 4.33.0 and Mathlib commit `db584cd6d46c92f209a44c0f1c829460d327499d`. Check the runtime with `lean --version`. From the package directory, `bash lean/verify.sh` runs:

```
lake build +Main
lake env lean -DwarningAsError=true Check.lean
lake env leanchecker --fresh Main
```

The dynamic check enumerates every owned namespace declaration, including generated declarations, in addition to the complete explicit-name list. Only `propext`, `Classical.choice`, and `Quot.sound` are allowed. Production timings are diagnostic and establish no controlled research-efficiency improvement.

Primary reference. M. Fathi and B. Nelson, *Free Stein kernels and an improvement of the free logarithmic Sobolev inequality*, arXiv:1605.05828v1, 19 May 2016: printed p.4 (difference quotients), p.7 (Definition 1.8), and pp.13–14 (the fixed unit-semicircle density and one-variable measure formula). `https://arxiv.org/abs/1605.05828v1`.